VENU. Good afternoon. I am Venu, with Bhuvan and Gowtham. We built a fuzzer, a dynamic-analysis tool that automatically generates test queries for PostgreSQL and decides by itself whether an answer is wrong. It found nine PostgreSQL bugs; six are fixed upstream, five with our patches. We then generalized it to four more SQL engines, where we filed three more fixes. We focus on how the tool works and how it found the bugs.

VENU. A fuzzer can run millions of queries; the hard part is to know that an answer is wrong. Our answer is comparison: equivalent questions must give equal answers.

VENU. Five steps. Derive a test case from our typed grammar; rewrite its trees into oracle checks; execute on master, under other planner settings and on 18.6; triage the failures. The LLM loop reads recent commits and proposes new grammar, which enters only if it passes a gate on the real server.

GOWTHAM. The grammar carries SQL types, so generated queries are valid and reach the planner. On top sits a stack of oracles. Three just watch the server. The others compare two answers that must agree. Because we keep the tree, a rewrite is type-correct at any nesting level.

BHUVAN. Triage makes a failure reportable: reproduce it, shrink it while keeping it valid by replaying trees, and find which setting or rewrite flips it. The LLM is never trusted on its word: PostgreSQL itself decides what enters the fuzzer.

GOWTHAM and BHUVAN. Read the middle column: every oracle earned its place. Two bugs came only from comparing versions, one only from the sub-select rewrite, one only from the hang oracle, one only from the estimate oracle. After the oracle fired, triage reduced and localized each case, and we wrote the root cause and the patch.

GOWTHAM. 5 trials of 1 hour per arm. Typing pays; oracles find real bugs, and their false positives were the main cost.

VENU. We then made the tool target-independent. Each engine gets a driver in its own worker process and a dialect overlay on the same grammar.

BHUVAN. Rule firing works on release builds of any engine and is about the optimizer, where logic bugs live. Normalizers make two engines comparable without hand-written rules.

VENU. Same pipeline as for PostgreSQL. The three pull requests were filed on 8 November and await review.

GOWTHAM. Valid SQL is the easy part; knowing the right answer is the hard part. Thank you.